Estd:1980

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
UG Programmes CE,CSE,ECE,EEE,IT & ME are Accredited by NBA
Chinna Amiram, Bhimavaram-534204. (AP)

Regulation: R20	I / IV - B.Tech. I - Semester								
COMPUTER SCIENCE & ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 HS 1101	English	HS	3	3	0	0	30	70	100
B20 BS 1101	Mathematics-I	BS	3	3	0	0	30	70	100
B20 BS 1103	Applied Chemistry	BS	3	3	0	0	30	70	100
B20 CS 1101	Programming for Problem Solving Using C	ES	3	3	0	0	30	70	100
B20 CS 1102	Computer Fundamentals and Digital Logic	ES	3	3	0	0	30	70	100
B20 CS 1103	Programming for Problem Solving Using C Lab	ES	1.5	0	0	3	15	35	50
B20 BS 1108	Applied Chemistry Lab	BS	1.5	0	0	3	15	35	50
B20 CS 1104	Computer Engineering Workshop	ES	1.5	0	0	3	15	35	50
		TOTAL	**19.5**	**15**	**0**	**9**	**195**	**455**	**650**

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20HS1101	**HS**	**3**	**--**	**--**	**3**	**30**	**70**	**3Hrs**

ENGLISH

(Common to AIDS,CE,CSE,ECE,EEE,IT&ME)

Introduction:

The course is designed to train students in receptive as well as productive skills by incorporating a comprehensive, coherent and integrated approach that improves the learners' ability to effectively use English language in academic/ workplace contexts. The shift is from *learning about the language* to *using the language*. On successful completion of the compulsory English language course/s in B.Tech., learners would be confident of appearing for international language qualification/proficiency tests such as GRE, GMAT, IELTS, TOEFL and BEC besides being able to handle the writing tasks and verbal ability components ofcampusplacementtests.Activitybasedteaching-learningmethodswouldbeadoptedto ensure that learners would engage in actual use of language both in the classroom and laboratorysessions.

Course Objectives:

1.	To facilitate effective listening skills for better comprehension of varied accents spoken at national and global levels.
2.	To focus on appropriate reading strategies for better comprehension of multiple texts and authentic materials.
3.	To improve speaking skills through participation in activities such as role plays, discussions and structured talks/oral presentations.
4.	To impart effective strategies for good writing and demonstrate the same in both summarizing and analyzing; writing well-organized essays, letters, e-mails, CV's and reports.
5.	To provide knowledge of grammatical structures and vocabulary and encourage their appropriate use in speech and writing.

Course Outcomes: At the end of the Course the students will be able to

S.No	OutCome	KL
1.	Identify the context, topic and pieces of specific information by understanding andresponding to the social or transactional dialogues spoken by native speakers of English.	K3
2.	Apply suitable strategies for skimming and scanning to get the main idea of a textand locate specific information.	K3
3.	Build confidence and adapt themselves to the social and public discourses,discussions and presentations.	K3
4.	Apply the principles of writing to paragraphs, arguments, essaysand formal/informal communication.	K3
5.	Construct sentences using proper grammatical structures and correct word forms.	K4

SYLLABUS

UNIT-I (8 Hrs)	**Lesson: A Drawer full of happiness** from *Infotech English*, Maruthi Publications. **Listening:** Listening to short audio texts and identifying the topic, context and specific pieces of information to answer a series of questions both in speaking and writing.

	Speaking: Self- introduction and introducing others. Asking and answering general questions on topics such as home, family, work, studies and interests. **Reading:** Skimming text to get the main idea. Scanning to look for specific pieces of information. **Reading for Writing:** Paragraph Writing (Hints Development), general essays using suitable cohesive devices; linkers, sign posts and transition signals; mechanics of writing, punctuation. **Vocabulary:**Technical vocabulary from across technical branches (20) GRE Vocabulary (20), antonyms and synonyms, word applications, verbal reasoning and sequencing of words. **Grammar:** Content words and function words; parts of Speech, tenses, word order in sentences, sentence structures.
UNIT-II (8 Hrs)	**Lesson-: Nehru's letter to his daughter, Indira on her birthday**from*Infotech English*, Maruthi Publications**.** **Listening:** Answering a series of questions about main idea and supporting ideas after listening to audio texts both in speaking and writing. **Speaking**: Discussion in pairs/ small groups on specific topics followed by short structured talks, functional English: greetings and leave takings. **Reading:** Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together. **Reading for Writing**: Identifying the main ideas, rephrasing and summarizing them (précis writing); avoiding redundancies and repetitions. **Vocabulary:** Technical vocabulary from across technical branches (20 words). GRE Vocabulary Analogies (20 words), antonyms and synonyms, word applications. **Grammar:** Articles, prepositions, conjunctions, use of synonyms and antonyms.
UNIT-III (8 Hrs)	**Lesson: Stephen Hawking-Positivity'Benchmark'**from*Infotech English*,Maruthi Publications. **Listening:**Listening for global comprehension and summarizing what is listened to both in speaking andwriting. **Speaking:** Discussing specific topics in pairs or small groups and reporting what is discussed. Functional English: complaining andapologizing. **Reading:** Reading a text in detail by making basic inferences -recognizing: and interpreting specific context clues; strategies to use text clues for comprehension, criticalreading. **Reading for Writing:** Letter writing- types, format and principles of letter writing, E-mail etiquette, writing a Resume/CV and covering letter. **Vocabulary:** Technical vocabulary from across technical branches (20 words. GRE. Vocabulary 20 words), Idioms & Phrasal verbs, Homonyms, word applications, sequencing of words.

	Grammar: Sentence Structures, Transformation of sentences (Active and passive Voice, Degrees of comparison, Simple, Compound and Complex).
UNIT-IV (8 Hrs)	**Lesson: Liking a Tree, Unbowed: Wangari Maathai biography**from *Infotech English*, MaruthiPublications. **Listening:** Making predictions while listening to conversations/ transactional dialogues without video (only audio), listening to audio-visual texts. **Speaking:** Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions. Functional English: asking for permissions, requesting,Inviting. **Reading:** Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicative process or display complicated data. **Reading for Writing:**Information transfer; describe, compare, contrast, identify significance/trends based on information provided in figures/charts/graphs/tables. Pamphlet writing, writing for media, writing SOP's. **Vocabulary:**Technical vocabulary from across technical branches (20 words GRE Vocabulary (20 words), antonyms and synonyms, word applications, cloze encounters, foreign phrases. **Grammar:**Quantifying expressions - adjectives and adverbs: comparing and contrasting, question Tags, direct and indirect speech, reporting for academic purposes.
UNIT-V (8 Hrs)	**Lesson: Stay Hungry–Stay Foolish**from *Infotech English*, Maruthi Publications. **Listening:** Identifying key terms, understanding concepts and interpreting the concepts both in speaking and writing. **Speaking:** Formal oral presentations on topics from academic contexts– with/without the use of PPT slides. Functional English: Suggesting/Opiniongiving. **Reading:** Reading for comprehension, RAP Strategy - intensive reading and extensive reading techniques. **Reading for Writing:** Report writing, writing academic proposals- writing research articles: format and style. **Vocabulary:** Technical vocabulary from across technical branches (20 words GRE 'Vocabulary (20 words, antonyms and synonyms, word applications, coherence, matching emotions. **Grammar:** Editing short texts — identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject-verb agreement, parallel structures, phrases and clauses).
Text Books:	
1	*Infotech English*, Maruthi Publications.

Reference Books:	
1.	Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge,2014.
2.	Chase. Becky Tarver. Pathways: Listening, Speaking and Critical Thinking. Heinley ELT;2nd Edition, 2018.
3.	Skilful Level 2 Reading & Writing Student's Book Pack (B1). Macmillan Educational.
4.	Hewing, Martin. Cambridge Academic English (B2). CUP, 2012.
E-Resources:	
Grammar/Listening/Writing	
	1-language.com
	http://www.5minuteenglish.com/
	https://www.englishpractice.com/
Grammar/Vocabulary	
	English Language Learning Online
	http://www.bbc.co.uk/learningenglish/
	http://www.better-english.com/
	http://www.nonstopenglish.com/
	https://www.vocabulary.com/
	BBC Vocabulary Games
	Free Rice Vocabulary Game
Reading	
	https://www.usingenglish.com/comprehension/
	https://www.englishclub.com/reading/short-stories.htm
	https://www.english-online.at/
Listening	
	https://learningenglish.voanews.com/z/3613
	http://www.englishmedialab.com/listening.html
Speaking	
	https://www.talkenglish.com/
	BBC Learning English – Pronunciation tips
	Merriam-Webster – Perfect pronunciation Exercises
All Skills	
	https://www.englishclub.com/
	http://www.world-english.org/
	http://learnenglish.britishcouncil.org/
	Online Dictionaries
	Cambridge dictionary online
	MacMillan dictionary
	Oxford learner's dictionaries

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1101	**BS**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

MATHEMATICS-I
(LINEAR ALGEBRA AND DIFFERENTIAL EQUATIONS)

(Common to AIDS, CE, CSE, ECE, EEE, IT & ME)

Pre-requisites: Calculus of functions of a single variable and Matrices.

Course Objectives: Students are expected to learn

1.	Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations.
2.	Eigen values, Eigen vectors and quadratic forms.
3.	First order ordinary differential equations and some simple geometrical and physical applications.
4.	Orthogonal trajectories, Simple electrical circuits and Newton's law of cooling.
5.	Methods of solution of linear higher order ordinary differential equations.
6	Concepts of Laplace transforms and their applications for solving ODE.

Course Outcomes: At the end of the course the student will be able to

S.No	Outcome	KL
1.	Solve a given system of linear algebraic equations	K3
2.	Determine Eigen values and Eigen vectors of a system represented by a matrix.	K3
3.	Solve ordinary differential equations of first order and first degree.	K3
4.	Applythe knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits	K3
5.	Solve linear ordinary differential equations of second order and higher order.	K3
6.	Determine Laplace transform, inverse Laplace transform and solve linear ODE	K3

SYLLABUS

UNIT-I (10 Hrs)	**Linear systems of equations:** Rank, Echelon form, Normal form, consistency of system of linear equations, Solution of linear systems by Gauss elimination, Jacobi and Gauss-Seidel methods.
UNIT-II (10 Hrs)	**Eigen values - Eigen vectors and Quadratic forms:** Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix using Cayley-Hamilton theorem, Reduction to diagonal form, Quadratic forms, Reduction of a Quadratic form to Canonical form.
UNIT-III (10 Hrs)	**Differential equations of first order and first degree:** Linear, Bernoulli, Exact, Reducible to exact types. Applications: Orthogonal trajectories, Newton's Law of cooling, Simple electrical circuits.(R-L and R-C circuits only)
UNIT-IV (8 Hrs)	**Linear differential equations of higher order:** Linear Non-homogeneous equations of higher order with constant coefficients with source (RHS) term of the type e^{ax}, sin ax, cos ax, polynomials in x, e^{ax} V(x), x V(x). Simultaneous differential equations with constant coefficients, Method of Variation of parameters.

UNIT-V (12 Hrs)	**Laplace transformation:** Laplace transforms of standard functions, properties,transforms of tf(t), f(t)/t, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function; Inverse Laplace transforms, convolution theorem (without proof). Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

Text Books:

1.	B.S.Grewal, Higher Engineering Mathematics, 43^{rd} Edition, Khanna Publishers.
2.	B. V. Ramana,Higher Engineering Mathematics, 2007 Edition, Tata Mc. Graw Hill Education.
3.	N.P.Bali&Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1.	V. Ravindranath&P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
2.	Erwin Kreyszig, Advanced Engineering Mathematics, 10^{th} Edition, Wiley-India.
3.	Michael Greenberg, Advanced Engineering Mathematics, 9^{th} edition, Pearson.
4.	Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
5.	Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
6.	Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
7.	Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1103	**BS**	**3**	**--**	**--**	**3**	**30**	**70**	**3Hrs**

APPLIED CHEMISTRY

(Common to CSE,ECE &IT)

Course Objectives:

1.	To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2.	To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3.	To understand the concept of galvanic cells and corrosion with theories like electro chemical theory.
4.	To understand the importance of water.
5.	To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes: At the end of the course students will be able to

S.no	Out Come	KL
1.	Develop polymer composites, synthetic polymers and formulation of polymers and their use in design	K3
2.	Apply the knowledge about quality of water and its treatment methods for domestic and industrial applications. Understanding the principle, mechanism of corrosion and utilization of various techniques to control.	K3
3.	Develop the knowledge of fuels and their economics, advantages and limitations. Make use of the basic concepts of semiconductors and liquid crystals for engineering applications.	K3
4.	Identify constituents of various ceramic materials, characteristics and their appropriate use in construction. Apply the knowledge of electrochemistry principles to design energy storage	K2

SYLLABUS

UNIT-I (10Hrs)	**High Polymers and Plastics; Rubbers & Elastomers** Polymerization Definition, Types of Polymerization, free radical Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;
UNIT-II (10Hrs)	**Energy Sources and Applications:** **Nuclear Energy:** Nuclear fission and Nuclear fusion – Nuclear Power reactor – Applications. Thermal fuels – Introduction – Classification – Calorific value – HCV and

	LCV – Bomb calorimeter; Coal : Proximate and ultimate analysis of coal – Significance of the analysis – Manufacture of coke by Otto Hoffman's by Product Process , Refining crude oil; Knocking; Chemical structure-Knocking, Octane number of gasoline, Cetane number of diesel oil, synthetic Petrol; LPG, CNG
UNIT-III (12Hrs)	**Electrochemical cells and Corrosion** Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen fuel cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion-material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electrrolessplating ;Paints
UNIT-IV (8Hrs)	**Water technology** Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.
UNIT-V (10Hrs)	**Chemistry of Engineering Materials& Advanced Engineering materials** **Cement:-** Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete. **Refractories:-** Definition, Characteristics, classification, Properties and failure of refactories. **Solar Energy: -** Construction and working of Photovoltaic cell, applications. **Solid State Materials:** Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors. **Liquid Crystals: -** Definition – Classification with examples – Applications

Text Books:	
1.	Engineering Chemistry by Jain and Jain, Dhanpat Rai Publishing co.
2.	Engineering Chemistry by Willy India Pvt Ltd.
3.	Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy ; Silicon Publications.

Reference Books:	
1.	Engineering Chemistry by Shikha Aharwal; Cambridge University Press, 2015 edition.
2.	A text of Engineering Chemistry by S.S.Dara; S.Chand& Co Ltd.
3.	Chemistry in Engineering and Technology by JC Kuriacose and J. Rajaram Mc. Graw Hill edition.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1101	ES	3	--	--	3	30	70	3 Hrs.

PROGRAMMING FOR PROBLEM SOLVING USING C

(Common to AIDS, CSE, ECE & IT)

Course Objectives:

1.	To learn about the computer systems, computing environments, developing of a computer program, Structure of a C Program and to evaluate expressions
2.	To gain knowledge of the operators, selection, control statements and repetition in C
3.	To learn about the design concepts of arrays, strings, enumerated structure and union types and their usage.
4.	To understand the concepts of pointers, dynamic memory allocation and know the significance of Preprocessor.
5.	To learn about various File I/O operations and significance of functions

Course Outcomes: At the end of the course the students will be able to

S.No	Outcome	KL
1.	Apply Precedence and Associativity rules to evaluate Expressions.	K3
2.	Make use of Decision Making and Looping statements to solve various problems in C	K3
3.	Illustrate the importance of Arrays and Strings and to apply various operations on them.	K2
4.	Solve various problems by making use of Structure and Union concepts	K3
5.	Design and implement programs to analyze the different pointer applications	K3
6.	Develop programs using Functions and Pointers.	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction to Computers: Creating and running Programs, Computer Numbering System, Storing Integers, Storing Real Numbers Introduction to the C Language: Background, C Programs, Identifiers, Types, Variable, Constants, Input/output, Programming Examples, Scope, Storage Classes and Type Qualifiers. Structure of a C Program: Expressions Precedence and Associativity, Side Effects, Evaluating Expressions, Type Conversion Statements, Simple Programs, Command Line Arguments.
UNIT-II (10 Hrs)	Bitwise Operators: Exact Size Integer Types, Logical Bitwise Operators, Shift Operators. Selection & Making Decisions: Logical Data and Operators, Two Way Selection, Multiway Selection, More Standard Functions. Repetition: Concept of Loop, Pretest and Post-test Loops, Initialization and Updating, Event and Counter Controlled Loops, Loops in C, Other Statements Related to Looping, Looping Applications, Programming Examples.
UNIT-III (10 Hrs)	Arrays: Concepts, Using Array in C, Array Application, Two Dimensional Arrays, Multidimensional Arrays, Programming Example – Calculate Averages Strings: String Concepts, C String, String Input / Output Functions, Arrays of Strings, String Manipulation Functions String/ Data Conversion, A Programming Example – Morse Code Enumerated, Structure, and Union: The Type Definition (Type def), Enumerated Types, Structure, Unions, and Programming Application.

UNIT-IV (10 Hrs)	Pointers: Introduction, Pointers to pointers, Compatibility, L value and R value Pointer Applications: Arrays, and Pointers, Pointer Arithmetic and Arrays, Memory Allocation Function, Array of Pointers, Programming Application. Processor Commands: Processor Commands.
UNIT-V (10 Hrs)	Functions: Designing, Structured Programs, Function in C, User Defined Functions, Inter Function Communication, Standard Functions, Passing Array to Functions, Passing Pointers to Functions, Recursion Text Input / Output: Files, Streams, Standard Library Input / Output Functions, Formatting Input / Output Functions, Character Input / Output Functions Binary Input / Output: Text versus Binary Streams, Standard Library, Functions for Files, Converting File Type.

Text Books:	
1.	Programming for Problem Solving, Behrouz A. Forouzan, Richard F.Gilberg, CENGAGE
2.	The C Programming Language, Brian W.Kernighan, Dennis M. Ritchie, 2e, Pearson
Reference Books:	
1.	Computer Fundamentals and Programming, Sumithabha Das, Mc Graw Hill.
2.	Programming in C, Ashok N. Kamthane, Amit Kamthane, Pearson.
3.	Computer Fundamentals and Programming in C, Pradip Dey, Manas Ghosh, OXFORD.
e-Resources:	
1.	https://www.geeksforgeeks.org/c-programming-language/
2.	https://www.learn-c.org/
3.	https://www.w3resource.com/c-programming-exercises/

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1102	**ES**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER FUNDAMENTALS AND DIGITAL LOGIC

(For CSE)

Course Objectives:

1.	To learn the computer fundamentals and internet of things concepts.
2.	To provide insights of various number systems, Boolean functions and logic gates.
3.	To introduce different minimization techniques.
4.	To design the various combinational and sequential circuits.

Course Outcomes: At the end of the course the students will be able to

S.No	Outcome	KL
1.	Familiar with computer fundamentals and internet of things concepts.	K1
2.	Differentiate binary, decimal, octal and hexadecimal number system.	K2
3.	Implement the Boolean functions using logic gates and Simplification Boolean expression using K-Map.	K3
4.	Implement various combinational circuits.	K3
5.	Implement various Sequential circuits.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Computer Fundamentals:** Basics of Introduction to Computer: History, Generations, Classifications of Computers, The Computer System Hardware: CPU, Memory, Input and Output Devices: Input Output Unit, Input Devices and Output devices, IO-Ports
UNIT-II (10 Hrs)	**Number Theory and BooleanAlgebra:** Binary Systems and Boolean Algebra Digital Systems. Binary Numbers. Number Base Conversions. Octal and Hexadecimal Numbers. Complements. Signed Binary Numbers. Binary Logic, Basic Definitions of Boolean algebra. Axiomatic Definition of Boolean Algebra. Basic Theorems and Properties of Boolean Algebra, Boolean Functions.
UNIT-III (10 Hrs)	**Logic Gates and Gate-Level Minimization Canonical and Standard Forms**. Logic Gates. The Map Method. Four-Variable Map. Five-Variable Map. Product of Sums Simplification. Don/t-Care Conditions.
UNIT-IV (10 Hrs)	**Combinational Logic Design:** Design Procedure, Binary Adder-Subtractor. Decimal Adder. Binary Comparator. Decoders. Encoders. Multiplexers.
UNIT-V (10 Hrs)	**Sequential Logic design Sequential Circuits**:Latches. Flip-Flops.Truth Tables. RS, JK, T and D Flip Flops, Truth and Excitation Tables, Conversion of Flip Flops. Analysis of Sequential Circuits. State Reduction and Assignment. Designs Procedure.

Text Books:

1.	Computer Fundamentals by Anitagoel Pearson education 2017
2.	Internet of things Architecture and Design Principles Rajkamal public. Mcgraw-hill education
3.	Digital design 5th edition by Morris Mano

Reference Books:	
1.	Switching and Finite Automata Theory by Zvi. Kohavi, Tata McGraw Hill.
2.	Switching and Logic Design, C.V.S. Rao, Pearson Education
3.	Digital Principles and Design – Donald D.Givone, Tata McGraw Hill, Edition.
4.	Fundamentals of Digital Logic & Micro Computer Design , 5TH Edition, M. Rafiquzzaman John Wiley

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1103	**ES**	**0**	**0**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

PROGRAMMING FOR PROBLEM SOLVING USING C LAB

(Common to AIDS, CSE, ECE & IT)

Course Objectives:

1.	Apply the principles of C language in problem solving.
2.	To design & develop of C programs using Arrays, Strings, Structures, Unions and Pointers
3.	To perform the file operations, preprocessor commands
4.	To solve various complex problem by applying modular programming skills

Course Outcomes: At the end of the course students will be able to

S.No	Out Come	KL
1.	Write, Trace and Debug the programs and correct syntax and logical errors.	K4
2.	Solve various Problems by making use of Arrays, Strings, Structures, Unions and Pointers	K3
3.	Solve a complex problem by decomposing into several modules by using Functions	K4
4.	Apply various File I/O operations	K3

LIST OF PROGRAMS

1	Exercise 1: 1. Write a C program to print a block F using hash (#), where the F has a height of six characters and width of five and four characters. 2. Write a C program to compute the perimeter and area of a rectangle with a height of 7 inches and width of 5 inches. 3. Write a C program to display multiple variables.
2	Exercise 2: 1. Write a C program to calculate the distance between the two points. 2. Write a C program that accepts 4 integers p, q, r, s from the user where r and s are positive and p is even. If q is greater than r and s is greater than p and if the sum of r and s is greater than the sum of p and q print "Correct values", otherwise print "Wrong values".
3	Exercise 3: 1. Write a C program to convert a string to a long integer. 2. Write a program in C which is a Menu-Driven Program to compute the area of the various geometrical shape. 3. Write a C program to calculate the factorial of a given number.
4	Exercise 4: 1. Write a program in C to display the n terms of even natural number and their sum. 2. Write a program in C to display the n terms of harmonic series and their sum. 1 + 1/2 + 1/3 + 1/4 + 1/5 ... 1/n terms. 3. Write a C program to check whether a given number is an Armstrong number or not.
5	Exercise 5: 1. Write a program in C to print all unique elements in an array. 2. Write a program in C to separate odd and even integers in separate arrays. 3. Write a program in C to sort elements of array in ascending order.

6	Exercise 6: 1. Write a program in C for multiplication of two square Matrices. 2. Write a program in C to find transpose of a given matrix.
7	Exercise 7: 1. Write a program in C to search an element in a row wise and column wise sorted matrix. 2. Write a program in C to print individual characters of string in reverse order.
8	Exercise 8: 1. Write a program in C to compare two strings without using string library functions. 2. Write a program in C to copy one string to another string.
9	Exercise 9: 1. Write a C Program to Store Information Using Structures with Dynamically Memory Allocation 2. Write a program in C to demonstrate how to handle the pointers in the program.
10	Exercise 10: 1. Write a program in C to demonstrate the use of & (address of) and *(value at address) operator. 2. Write a program in C to add two numbers using pointers
11	Exercise 11: 1. Write a program in C to add numbers using call by reference. 2. Write a program in C to find the largest element using Dynamic Memory Allocation
12	Exercise 12: 1. Write a program in C to swap elements using call by reference. 2. Write a program in C to count the number of vowels and consonants in a string using a pointer.
13	Exercise 13: 1. Write a program in C to show how a function returning pointer. 2. Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc( ) function
14	Exercise 14: 1. Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc( ) function. Understand the difference between the above two programs 2. Write a program in C to convert decimal number to binary number using the function.
15	Exercise 15: 1. Write a program in C to check whether a number is a prime number or not using the function. 2. Write a program in C to get the largest element of an array using the function.
16.	Exercise 16: 1. Write a program in C to append multiple lines at the end of a text file. 2. Write a program in C to copy a file in another name. 3. Write a program in C to remove a file from the disk.

Reference Books:

1.	Programming for Problem Solving, Behrouz A. Forouzan, Richard F.Gilberg, CENGAGE
2.	The C Programming Language, Brian W.Kernighan, Dennis M. Ritchie, 2e, Pearson

e-Resources:

1.	https://www.geeksforgeeks.org/c-programming-language/
2.	https://www.learn-c.org/
3.	https://www.tutorialspoint.com/cprogramming/index.html

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1108	BS	--	--	3	1.5	15	35	3Hrs

APPLIED CHEMISTRY LAB

(Common to CSE,ECE &IT)

Course Objectives:

1.	To investigate and understand Physical behaviour in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2.	Effectively communicate experimental results and solutions to application problems through oral and written reports.
3.	Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes: At the end of the course students will be able to

S.No	Out Come	KL
1.	Gain technical knowledge of measuring, operating and testing of chemical instruments and equipments. Carrying out different types of chemical reactions for analysing different materials in micro level quantities.	K3
2.	Analyze and generate experimental skills to enhance the analytical thinking capabilities in the modern trends in engineering and technology.	K3

LIST OF PROGRAMS

1	Determination of Alkalinity of water sample.
2	Determination of total hardness of water by EDTA method.
3	Estimation of Ferrous Iron by $KMnO_4$.
4	Estimation of oxalic acid by $KMnO_4$.
5	Estimation of Mohr's salt by $K_2Cr_2O_7$.
6	Estimation of Dissolved oxygen by Winkler's method.
7	Determination of pH of water and soil sample.
8	Determination of Chlorides present in water sample.
9	Conductometric titration of strong acid Vs strong base.
10	Potentiometric titration of strong acid Vs strong base.
11	Potentiometric titration of strong acid Vs weak base.
12	Preparation of Phenol formaldehyde resin.
13	Determination of saponification value of oils.
14	Determination of pour and cloud points of lubricating oil.
15	Determination of Acid value of oil.

Reference Books:

1.	Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.EngineeringCollege.
2.	Laboratory manual on Engineering Chemistry by Dr.Sudha Rani; Dhanpat Rai Publishing Company.
3.	Engineering Chemistry Laboratory manual – I & II by Dr.K.Anji Reddy; Tulip Publications.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1104	**ES**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

COMPUTER ENGINEERING WORKSHOP

(Common to AIDS, CSE & IT)

Course Objectives: Skills and knowledge provided by this subject are the following:

1.	PC Hardware: Identification of basic peripherals, Assembling a PC, Installation of system software like MS Windows, device drivers, etc. Troubleshooting of PC Hardware and Software issues.
2.	Internet & World Wide Web: Different ways of hooking the PC on to the internet from home and workplace and effectively usage of the internet, web browsers, email, newsgroups and discussion forums. Awareness of cyber hygiene (protecting the personal computer from getting infected with the viruses), worms and other cyber attacks.
3.	Productivity Tools: Understanding and practical approach of professional word documents, excel spread sheets, power point presentations and personal web sites using the Microsoft suite office tools.

Course Outcomes

S.No	Students will be able to	KL
1	Identify, assemble and update the components of a computer	K3
2	Configure, evaluate and select hardware platforms for the implementation and execution of computer applications, services and systems	K3
3	Make use of tools for converting pdf to word and vice versa	K3
4	Develop presentation, documents and small applications using productivity tools such as word processor, presentation tools, spreadsheets, HTML, LaTex	K3

SYLLABUS

Note: Faculty to consolidate the workshop manuals using the textbook and references

List of Exercises:

Task 1	**Identification of the peripherals of a computer -** Prepare a report containing the block diagram of the computer along with the configuration of each component and its functionality. Describe about various I/O Devices and its usage.
Task 2:	Practicing disassembling and assembling components of a PC
Task 3:	Installation of Device Drivers, MS Windows, Linux Operating systems and Disk Partitioning, dual boating with Windows and Linux
Task 4:	Introduction to Memory and Storage Devices, I/O Port, Assemblers, Compilers, Interpreters, Linkers and Loaders.
Task 5:	Demonstration of Hardware and Software Troubleshooting
Task 6:	Demonstrating Importance of Networking, Transmission Media, Networking Devices-Gateway, Routers, Hub, Bridge, NIC, Bluetooth Technology, Wireless Technology, Modem, DSL, and Dialup Connection.
Task 7	Surfing the Web using Web Browsers, Awareness of various threats on the Internet and its

	solutions, Search engines and usage of various search engines, Need of anti-virus, Installation of anti-virus, configuring personal firewall and windows update. (Students should get connected to their Local Area Network and access the Internet. In the process they should configure the TCP/IP setting and demonstrate how to access the websites and email. Students customize their web browsers using bookmarks, search toolbars and pop up blockers)
Task 8:	**Productivity Tools:** Basic HTML tags, Introduction to HTML5 and its tags, Introduction to CSS3 and its properties. Preparation of a simple website/ homepage, Assignment: Develop your home page using HTML Consisting of your photo, name, address and education details as a table and your skill set as a list. Features to be covered:- Layouts, Inserting text objects, Editing text objects, Inserting Tables, Working with menu objects, Inserting pages, Hyper linking, Renaming, deleting, modifying pages,etc.,
Task 9:	Demonstration and Practice of various features of Microsoft Word Assignment: 1. Create a project certificate. 2.Creating a newsletter Features to be covered:-Formatting Fonts, Paragraphs, Text effects, Spacing, Borders and Colors, Header and Footer, Date and Time option, tables, Images, Bullets and Numbering, Table of Content, Newspaper columns, Drawing toolbar and Word Art and Mail Merge in word etc.,
Task 10:	Demonstration and Practice of various features MicrosoftExcel Assignment: 1. Creating a scheduler 2.CalculatingGPA 3.Calculating Total, average of marks invarious subjects and ranks of Students based on marks Features to be covered:- Format Cells, Summation, auto fill,Formatting Text, Cell Referencing, Formulae in excel, Charts, Renaming and Inserting worksheets,etc.,
Task 11:	Demonstration and Practice of various features Microsoft Power Point Features to be covered:- Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Hyperlinks Tables and Charts, Master Layouts, Types of views, Inserting – Background, textures, Design Templates, etc.,
Task 12:	Demonstration and Practice of various features LaTeX – document preparation, presentation (Features covered in Task 9 and Task 11 need to be explored in LaTex)
Task 13:	Tools for converting word to pdf and pdf to word
Task14:	Internet of Things (IoT): IoT fundamentals, applications, protocols, communication models, architecture, IoT devices.

Text Books:

1	Computer Fundamentals, Anita Goel, Pearson India Education,2017
2	PC Hardware Trouble Shooting Made Easy,TMH
3	Introduction to Information Technology, ITL Education Solutions Limited,2nd Edition, Perason, 2020
4	Upgrading and Repairing PCs, 18th Edition, Scott Mueller, QUE, Pearson,2008

5	LaTeX Companion – Leslie Lamport,PHI/Pearson
6	Introducing HTML5, Bruce Lawson, Remy Sharp, 2nd Edition, Pearson,2012
7	Teach yourself HTML in 24 hours, ByTechmedia
8	HTML 5 and CSS 3.0 to the Real World by Alexis Goldstein, Sitepointpublication.
9	Internet of Things, Technologies, Applications, Challenges and Solutions, B K Tripathy, J Anuradha, CRCPress
10	Comdex Information Technology Course Tool Kit, Vikas Gupta, WileyDreamtech.
11	IT Essentials PC Hardware and Software Companion Guide Third Edition by David Anfinson and Ken Quamme, CISCO Press, PearsonEducation.
12	Essential Computer and IT Fundamentals for Engineering and Science Students,Dr. N. B. Venkateswarlu, S. Chand Publishers

Regulation: R20	I / IV - B.Tech. II - Semester								
COMPUTER SCIENCE & ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 1201	Mathematics-II	BS	3	3	0	0	30	70	100
B20 BS 1202	Applied Physics	BS	3	3	0	0	30	70	100
B20 BS 1204	Mathematical Foundations of Computer Science	ES	3	3	0	0	30	70	100
B20 CS 1202	Computer Organization	ES	3	3	0	0	30	70	100
B20 CS 1203	Data Structures	ES	3	3	0	0	30	70	100
B20 BS 1207	Applied Physics Lab	BS	1.5	0	0	3	15	35	50
B20 HS 1202	Communication Skills Lab	HS	1.5	0	0	3	15	35	50
B20 CS 1206	Data Structures Lab	ES	1.5	0	0	3	15	35	50
B20 MC 1202	Professional Ethics and Human Values	MC	0	2	0	0	--	--	--
B20 MC 1203	National Service Scheme (NSS)	MC	0	0	0	2	--	--	--
		TOTAL	**19.5**	**17**	**0**	**11**	**195**	**455**	**650**

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1201	BS	3	--	--	3	30	70	3 Hrs.

MATHEMATICS – II
(FOURIER ANALYSIS AND PARTIAL DIFFERENTIAL EQUATIONS)

(Common to AIDS, CE, CSE, ECE, EEE, IT & ME)

Prerequisites: Calculus of functions of a single variable and Geometry

Course Objectives: Students are expected to learn:

1.	How to expand an aperiodic function in a Fourier series.
2.	How to find Fourier transform for a given function and evaluate some real definite integrals.
3.	Application of partial differentiation for determining maxima/ minima of functions.
4.	Evaluation of real definite integrals.
5.	Formation and solution of linear partial differential equations
6.	Solution of one-dimensional wave equation and one-dimensional heat equation by the method of separation of variables.

Course Outcomes: At the end of the course students will be able to

S. No	Outcome	KL
1.	DetermineFourier series and half range series of functions	K3
2.	Determine Fouriertransforms of non-periodic functions and also use them to evaluate integrals.	K3
3.	Compute partial derivatives, total derivative and Jacobians.	K3
4.	Find maxima/minima of functions of two variables and evaluate some real definite integrals.	K3
5.	Form partial differential equations and solve Lagrange linear equation. Solve linear higher order homogeneous and non-homogeneous PDEs.	K3
6.	Find theoretical solution of one-dimensional wave equation and one-dimensional heat equation	K3

SYLLABUS

UNIT-I (10 Hrs)	**Fourier Series** Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Change of interval. Even and odd functions, Half-range sine and cosine series.
UNIT-II (12 Hrs)	**Fourier Transforms** Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, Finite Fourier transforms, properties, inverse transforms, Parseval's Identities.
UNIT-III (10 Hrs)	**Partial differentiation:** Introduction, Homogeneous functions, Euler's theorem, Chain rule, Total derivative, Jacobians and their properties. Applications: Taylor series expansion for a function of two variables, Maxima and Minima of functions of two variables with and without constraints, Lagrange's method.Leibnitz's rules for differentiation under integral sign.

UNIT-IV (10 Hrs)	**First order and higher order partial differential equations:** Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of Lagrange linear equation.Solutions of Linear homogeneous and non-homogeneous partial differential equations with constant coefficients –source (RHS) terms of the type e^{ax+by}, sin (ax+by), cos (ax+by), $x^m y^n$.
UNIT-V (10 Hrs)	**Applications of partial differential equations:** Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one- dimensional heat equation

Text Books:

1.	B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2.	N.P.Bali& Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.
3.	B. V. Ramana, Higher Engineering Mathematics, 2007 Edition, Tata Mc. Graw Hill Education.

Reference Books:

1.	Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2.	V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3.	Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4.	David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5.	Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6.	Dass H.K., RajnishVerma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1202	BS	3	--	--	3	30	70	3 Hrs.

APPLIED PHYSICS

(Common to CSE, ECE &IT)

Course Objectives:

1.	**Impart** the knowledge in basic concepts of wave optics through the Phenomena of interference and diffraction, basic concepts and properties of dielectric and magnetic materials and semiconductors.
2.	**Familiarize** the student with modern technologies like lasers, optical fibers and ultrasonics with an understanding of the science behind.
3.	**Impart** the elementary concepts of nanomaterials and their significance in different engineering branches.

Course Outcomes: At the end of the course the student will be able to

S.No	Outcome	KL
1.	**Interpret** the behavior of light radiation in interference and diffraction Phenomena and their applications.	K3
2.	**Explain** the classification and properties of dielectric and magnetic materials suitable for engineering applications.	K3
3.	**Understand** the basics of modern optical technologies like lasers and optical fibers and their utility in various fields.	K3
4.	**Explain** the important aspects of semiconductors and electrical conductivity in them.	K3
5.	**Understand** the basics of technology of Ultrasonics in various fields and demonstrate the synthesis and applications of nanomaterials.	K3

SYLLABUS

UNIT-I (10 Hrs)	**WAVE OPTICS** **Interference:** Principle of super position. Interference of light, interference in thin films (reflected light) – Wedge film and Newton`s rings – Applications **Diffraction:** Types of diffraction, Fraunhoffer diffraction at a single alit, Diffraction grating, grating spectrum. Missing order, Resolving power, Rayleigh's Criterion, Resolving power of Grating
UNIT-II (10 Hrs)	**DIELECTRICS AND MAGNETICS** **Dielectrics** : Introduction to dielectrics, Electric Polarization, Dielectric polarizability, Susceptibility, Dielectric constant, Types of Polarization, Frequency dependence of Polarization, Internal field in a dielectric, Claussius and Mosotti equation, Applications of dielectrics. **Magnetics**: Introduction to magnetics, Magnetic dipole moment , Magnetization, Magnetic susceptibility and Permeability, Origin of permanent magnetic moment, Classification of magnetic materials (Dia , Para, Ferro, Antiferro and ferri), Hysteresis – Weiss Domain theory – Ferrites, soft and hard magnetic materials, Magnetic device applications.

UNIT-III (10 Hrs)	**LASERS AND FIBER OPTICS** **Lasers**: Introduction, Interaction of radiation with matter, condition for light amplification, Einstein's relations. Requirements of lasers device Types of lasers, Design and working of Ruby and He – Ne lasers, Laser characteristics and applications. **Fiber Optics**: Introduction to optical fibers, Principle of light propagation in fiber, Acceptance angle, Numerical aperture, Modes of propagations, types of fibers, classification of fibers based on refractive index profile, applications of fibers with emphasis on fiber optic communication.
UNIT-IV (9 Hrs)	**SEMICONDUCTORS** Introduction, intrinsic semi conductors, density of charge carries, Fermi energy, Electrical conductivity – Extrinsic semi conductors – P-type and N-type, Density of charge carriers, dependence of Fermi energy on carrier concentration and temperature, direct and indirect band – gap semi conductors, Hall effect, Applications of Hall effect. Drift and diffusion currents, Continuity equation, applications of semi conductors.
UNIT-V (9 Hrs)	**ULTRASONICS AND NANOMATERIALS** **Ultrasonics:** Introduction, Production of Ultrasonics – Piezoelectric and Magnetostriction methods, detection of ultrasonics, acoustic grating - determination of wavelength and velocity of ultrasonics, applications of ultrasonics. **Nanomaterials:** Introduction, salient features of Nanomaterials, Synthesis methods – Ball milling, Condensation, Chemical Vapour Deposition and Sol – Gel methods, Characterization techniques for nano materials - The scanning tunneling microscopy (STM) and The atomic force microscopy (AFM), Carbon nanotubes (CNTS), Applications of Nano materials.

Text Books:	
1.	A text Book of Engineering Physics – M.N. Avadhanulu and P.G.Kshirasagar.-S.Chand Publications 2017
2.	Engineering Physics by HK Malik and A.K.Singh. McGrawhill Publishing Company Ltd.
3.	Engineering Physics by V.Rajendran. McGrawhill Education (India)Pvt Ltd.
Reference Books:	
1.	Introduction to Solid State Physics by Charles Kittel , Wiley Publications 2011
2.	Semiconductors Devices – Physics and Technology by S.M.Sze , Wiley Publications 2008
3.	Text book of Nano Science and Nano technology by TataMcGrawhill 2013.
4.	Optical fiber communications by Gerd Keiser, Tata McGraw hill 2008.
e-Resources:	
1.	http://library.iiti.ac.in/
2.	https://onlinecourses.nptel.ac.in/

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1204	**BS**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

(For CSE)

Course Objectives: Students are expected to

1.	Understand propositional and predicate calculus.
2.	Know about concepts of counting techniques.
3.	Identify various types of relations and discuss their properties.
4.	Understand the concepts in Lattices and Boolean Algebra.
5.	Know about generating functions and methods of solving recurrence relations
6.	Have an idea on the concepts of Graph theory & Tree structures

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	KL
1.	Write and verify the arguments for their validity using propositional and predicate logic.	K3
2.	Utilize different counting methods in their fields of study.	K3
3.	Make use of various types of relations and their properties.	K3
4.	Identify different Lattices and Boolean expressions.	K3
5.	Formulate and solve the recurrence relations.	K3
6.	Utilize the concepts in graphs and trees.	K3

SYLLABUS

UNIT-I (12 Hrs)	**Mathematical Logic:** Propositional Calculus: Statements and Notations, Connectives, Well-formed Formulae, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.
UNIT-II (08 Hrs)	**Combinatorics**: Basics of Counting, Permutations, Permutations with Repetitions, Circular Permutations, Restricted Permutations, Combinations, Restricted Combinations, Generating Functions of Permutations and Combinations, Binomial and Multinomial Theorems, Binomial and Multinomial Coefficients, Principle of Inclusion–Exclusion.
UNIT-III (14 Hrs)	**Relations, Lattices & Boolean Algebra:** **Relations :** Definition of Relation, Properties of Binary Relations, Relation matrix and diagraph, Operations on Relations, Transitive Closure, Warshall’s algorithm, Equivalence and Compatibility relations, Partial Ordering Relations, Hasse Diagrams. **Lattices & Boolean Algebra:** Lattices and their properties, different types of lattices, Boolean algebra- Boolean expressions, truth tables and karnaugh maps

UNIT-IV (10 Hrs)	**Recurrence Relations**: Generating Functions, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations
UNIT-V (12 Hrs)	**Graph Theory:** Basic Concepts of Graphs, Sub graphs, Isomorphism of Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Multigraphs, Bipartite graphs, Planar Graphs, Euler's Formula. **Trees**: Definition of Tree, properties of Trees, Different tree structures, Binary trees, Spanning trees, Minimal Spanning Trees, Kruskal's and Prim's Algorithms.

Text Books:

1.	Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2.	Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India

Reference Books:

1.	Elements of Discrete Mathematics-A Computer Oriented Approach, C. L. Liu and D.P. Mahopatra, 3rd Edition, Tata McGraw Hill.
2.	Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
3.	Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby, Sharon Cutler Ross, PHI.
4.	Discrete Mathematics, S. K. Chakraborthy and B.K. Sarkar, Oxford, 2011.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1202	**ES**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER ORGANIZATION

(For CSE)

Course Objectives:

1.	Learn basic building blocks of a computer and their organization.
2.	Design a basic computer.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	KL
1.	Identify basic building blocks of a computer.	K2
2.	Design of computer functional blocks.	K6
3.	Identify Regular operation of a computer	K3
4	Identify the parameters that enhance system performance.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Digital Computers and Arithmetic** Historical perspective and von Neumann computers, Fixed and floating-point representation of numbers, Addition and Subtraction, Multiplication and Division algorithms, Floating-point arithmetic operations.
UNIT-II (10 Hrs)	**Basic Computer Organization and Design:** Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Input Output and Interrupt, Complete Computer Description, Design of Basic Computer. **Micro programmed Control:** Control Memory, Address Sequencing, Micro program Example, Design of Control Unit
UNIT-III (10 Hrs)	**Central Processing Unit:** Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer(RISC)
UNIT-IV (10 Hrs)	**Memory and I/O Organization** Memory Hierarchy, Associative Memory, Cache Memory, Virtual memory. I/O Organization: Peripheral devices, I/O interface, Asynchronous data transfer, Modes of transfer, Priority interrupt, direct memory access and IOP
UNIT-V (10 Hrs)	**Pipeline and Vector Processing** Parallel Processing, Pipelining, Arithmetic and Instruction Pipelines, RISK Pipeline, Vector Processing, Array Processors. Multiprocessors, Interconnection structures, Cache coherence.

Text Books:	
1.	Computer System Architecture, M. Morris Mano, Prentice Hall of India Pvt. Ltd., Third Edition, Sept.2008.
Reference Books:	
1.	Computer Organization and Architecture-Designing for Performance, William Stallings, Pearson, 9th ed.,2013
2.	Essentials of Computer Organization and Architecture, Linda Null, JuliaLobur, Narosa Pub., 3rd ed.,2003,
3.	Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat ZakyandZvonko Vranesic, 5th ed., TMH, 2011.
4.	Computer System Architecture", John. P. Hayes.
5.	Computer Systems Organization & Architecture, JohnD.Carpinelli, Addison Wesley, 2001.
6.	Computer Organization, Design and Architecture, SajjanG.Shiva, 4thed. CRCPress, 2008.
7.	Fundamentals of Computer Organization and Design, Sivarama P. Dandamudi, Springer-Verlag,2003
8.	Computer Architecture and organization: An Integrated Approach, Miles Murdocca and Vincent Heuring, Wiley, 2007.
9.	Computer Organization and Architecture: Themes and Variations, Alan Clements, Cengage Learning, 2014.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1203	**ES**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DATA STRUCTURES

(For CSE)

Course Objectives:

1.	Be familiar with basic techniques of algorithm analysis
2.	Master the implementation of data structures like stacks, queues, linked lists, binary trees, graphs.
3.	Be familiar with basic techniques for algorithm development like recursion.
4.	Be familiar with several sub-quadratic sorting algorithms including quick sort, merge sort and heap sort
5.	Master analyzing problems and writing program solutions to problems using the above techniques.

Course Outcomes:

S.No	Outcome	KL
1.	Demonstrate the concept of recursion, the way arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory.	K3
2.	Implement stacks, linked lists, queues and trees and apply them to solve different Computer Science problems and Engineering problems.	K3
3.	Compare alternative implementations of data structures with respect to performance.	K4
4.	Apply the principal algorithms for sorting and searching to the given data and analyze the computational efficiency.	K3
5.	Make use of Graphs to solve real life applications.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Basic Concepts: Arrays, Structures:** System Life Cycle, Algorithm Specification, Data Abstraction, Performance Analysis, Space Complexity, Time Complexity, Asymptotic Notation, Comparing Time Complexities. Array as an Abstract Data Type, Polynomial Abstract Data Type, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures **Simple Searching and Sorting Techniques:** Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Shell Sort, Introduction to Merge Sort, **Introduction to Recursion**: Towers of Hanoi, Quick Sort, Merge Sort, Complexity Analysis of Basic Sorting and Searching techniques
UNIT-II (10 Hrs)	**Stacks, Queues** Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Circular Queues using arrays. Pointers, Dynamically Allocated Storage using pointers, Dynamically Linked Stacks and Queues
UNIT-III (10 Hrs)	**Linked Lists:** **Singly Linked Lists**: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from linked list, Radix Sort.

	Circular Linked Lists: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from Circular Linked Lists. **Doubly Linked Lists**: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from Doubly Linked Lists. **Polynomials**: Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials.
UNIT-IV (10 Hrs)	**Trees:** Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree.
UNIT-V (10 Hrs)	**Graphs:** Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm
Text Books:	
1.	Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.
Reference Books:	
1.	Data Structures using C by Aaron M. Tenenbaum,Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2.	Data Structures with C by Seymour lipschutz, Schaum Outline series, 2010.
3.	Data Structures using C by R. KrishnaMoorthy G. Indirani Kumaravel, TMH, New Delhi,2008.
e-Resources	
1.	https://nptel.ac.in/courses/106/102/106102064/
2.	https://www.tutorialspoint.com/data_structures_algorithms/index.htm
3.	https://www.geeksforgeeks.org/data-structures/

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS1207	BS	--	--	3	1.5	15	35	3 Hrs.

APPLIED PHYSICS LAB

(Common to CSE,ECE &IT)

Course Objectives:

1.	To impart hands-on experience to the students entering engineering / Technology education about handling sophisticated equipment / instruments.
2.	To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes: At the end of the course students willbe able to

S.No	**Out Come**	**KL**
1.	Get hands on experience in setting up experiments and using the instruments / equipment individually.	K3
2.	Get introduced to using new / advanced technologies and understand their significance.	K3

LIST OF EXPERIMENTS

1	Determination of the Wavelength of light from a source – Diffraction Grating – Normal incidence.
2	Determination of radius of curvature of Plano convex lens – Newton's Rings.
3	Determination of the thickness of a thin spacer using interference – Air Wedge method.
4	Determination of Magnetic field along the axis of a current carrying coil –Stewart and Gee's apparatus.
5	Verification of Laws of series and parallel combinations of resistances – Carey Foster's bridge.
6	Determination of Temperature Coefficient of Resistance of a thermistor
7	To study the characteristics of PN Junction diode
8	To determine the Numerical aperture of a given optical fiber and hence to find its acceptance angle.
9	Determination of Planck constant
10	Determination of the Rigidity modulus of elasticity of a material – Torsional pendulum.
11	Verification of the laws of vibrations in stretched stings - Sonometer.
12	Determination of the frequency of the AC supply – AC Sonometer.
13	To determine refractive indices (μo and μe) of a birefringent material (prism).

Reference Books:

1. Advanced Practical Physics Vol 1& 2 SP Singh & M.S Chauhan Pragati Prakashan ,Meerut

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20HS1202	**HS**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3Hrs**

COMMUNICATION SKILLS LAB

(Common to AIDS ,CE,CSE,ECE,EEE,IT & ME)

Course Objectives:

1.	To expose to a variety of self-instructional, learner-friendly modes of language learning.
2.	To familiarize the students with CALL (Computer Assisted Language Learning). Thus, providing them with the required facility to face computer-based competitive exams like GRE,TOEFL, GMAT etc.
3.	To equip the students with necessary professional communication.
4.	To build confidence in LSRW Skills.
5.	To adapt the students by adopting the techniques of effective communication skills.

Course Outcomes: At the end of the course students will be able to

S.No	Out Come	KL
1.	Apply their linguistic competence in all LSRW skills to professional and personal settings.	K3
2.	Apply communication skills learnt through various language learning activities to their advancement in academics and competitive examinations.	K3
3.	Draft job application letters, E-Mail messages and other writing discourses.	K3
4.	Adopt professional etiquette consistent with formal settings.	K3
5.	Improve fluency and clarity in both spoken and written English.	K3

SYLLABUS

UNIT-I	A list of communicative expressions (Requests, Permissions, Asking/ giving directions, Thanking and Responding to Thanks, Clarifying, Inviting, Congratulating, Advising, Agreeing and disagreeing etc.,) Common Errors
UNIT-II	Pronunciation Letters and Sounds The Sounds of English Stress and Intonation Phonetic Transcription
UNIT-III	Group Discussions
UNIT-IV	Presentation Skills
UNIT-V	Interview Skills Resume/ Curriculum Vitae Covering Letter FAQ's Telephonic Interviews/ Etiquette Mock Interviews

Text Books:	
1.	Interact – English Lab Manual for Undergraduate Students – Orient BlackSwan
Reference Books:	
1.	Exercises in Spoken English Part 1,2,3,4, OUP and CIEFI.
2.	English Pronunciation in use- Mark Hancock, CUP.
3.	English Pronunciation in use- Mark Hewings, CUP.
4.	English Pronunciation Dictionary- Daniel Jones, CUP.
5.	English Phonetics for Indian Students- P. BalaSubramanian, Mac MillanPublications
6.	Technical Communication- Meenakshi Raman, Sangeeta Sharma, OUP.
7.	Technical Communication- Gajendra Singh Chauhan, SmitaKashiramka, cengage Publications

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS1206	**ES**	**0**	**0**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

DATA STRUCTURES LAB

(For CSE)

Course Objectives:

1.	To implement stacks and queues using arrays and linked lists.
2.	To develop programs for searching and sorting algorithms.
3.	To write programs using concepts of various trees.
4.	To implement programs using graphs.

Course Outcomes:

1.	Student will be able to write programs to implement stacks and queues.	K3
2.	Ability to implement various searching and sorting techniques.	K4
3.	Ability to implement programs using trees and graphs.	K4

LIST OF PROGRAMS

1	Write a program for sorting a list using Bubble sort and then apply binary search.
2	Write a program to implement the operations on stacks.
3	Write a program to implement the operations on circular queues.
4	Write a program for evaluating a given postfix expression using stack.
5	Write a program for converting a given infix expression to postfix form using stack.
6	Write a program for implement the following using recursion i) Towers of Hanoi ii) GCD of two numbers iii) Maximum element in an array
7.	Write a program to implement insert, delete, traverse, search operations on singly linked lists
8.	Write a program to implement insert, delete, traverse, search operations on circular linked lists
9.	Write a program to implement insert, delete, traverse, search operations on doubly linked lists
10	Write a program for the representation of polynomials using linked list and for the addition of two such polynomials.
11	Write a program for quick sort .
12	Write a program for Merge sort.
13	Write a program for Heap sort .
14	Write a program to create a binary search tree and for implementing the in order, preorder, post order traversal using recursion.
15	Write a program for finding the transitive closure of a digraph.
16	Write a program for finding the shortest path from a given source to any vertex in a digraph using Dijkstra’s algorithm.
17	a)Write a program for finding the Depth First Search of a graph. b)Write a program for finding the Breadth First Search of a graph
18	Write a program to implement Prims Algorithm.

Reference Books:	
1.	Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani and Anderson-Freed, Universities Press, 2008.
e-Resources	
1	https://nptel.ac.in/courses/106/102/106102064/
2	https://www.tutorialspoint.com/data_structures_algorithms/index.htm
3	https://www.geeksforgeeks.org/data-structures/

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC1202	MC	2	--	--	0	--	--	--

PROFESSIONAL ETHICS AND HUMAN VALUES

(Common to CSE, ECE & IT)

Course Objectives:

1	To create an awareness on Engineering Ethics and Human Values.
2	To instill Moral and Social Values and Loyalty.
3	To appreciate the rights of others.
4	To create awareness on assessment of safety and risk.

	Course Outcomes: At the end of the course students will be able to:	K L
1	Identify and analyze an ethical issue in the subject matter under investigation or in a relevant field. Demonstrate knowledge of ethical values in non-classroom activities, such as service learning, internships and field work.	K1&K2
2	Identifythemultipleethicalinterestsatstakeinareal-worldsituationorpractice and Articulatewhat makesaparticularcourseofactionethicallydefensible.	K1&K2
3	Assess their own ethical values and the social context of problems.	K3
4	Identify ethical concerns in research and intellectual contexts, including academic integrity,useandcitationofsources,theobjectivepresentationofdata,andthetreatment of humansubjects.	K3
5	Integrate, synthesize, and apply knowledge of ethical dilemmas and resolutions in academic settings, including focused and interdisciplinary research.	K4

SYLLABUS

UNIT-I (10 Hrs)	**Human Values:** Morals,ValuesandEthics-Integrity-WorkEthic-ServicelearningCivicVirtue Respect for others Living Peacefully Caring Sharing Honesty -Courage-Cooperation Commitment Empathy Self Confidence CharacterSpirituality.
UNIT-II (10 Hrs)	**Engineering Ethics:** Senses of 'Engineering Ethics-Variety of moral issued- Types of inquiry Moral dilemmas Moral autonomy- Kohlberg's theory- Gilligan's theory -Consensus and controversy Models of professional roles-Theories about right action-Self-interest - Customs and religion Uses of Ethical theories Valuing time Cooperation Commitment.
UNIT-III (8 Hrs)	**Engineering as Social Experimentation:** Engineering As SocialExperimentation- Framing the problem- Determiningthefacts codesofEthics- ClarifyingConcepts- Application issues Common Ground -General Principles- Utilitarian thinking respect forpersons.
UNIT-IV (10 Hrs)	**Engineers Responsibility for Safety and Risk:** Safety and risk Assessment of safety and risk. Risk benefit analysis and reducing risk- Safety and the Engineer-Designing for the safety- Intellectual Property rights(IPR).

UNIT-V (10Hrs)	**Global Issues:** Globalization- Cross-culture issues-Environmental Ethics- Computer Ethics Computers as the instrument of Unethical behavior Computers as the object of Unethical acts Autonomous Computers-Computer codes of Ethics- Weapons Development -Ethics and Research Analyzing Ethical Problems in research.

Text Books:

1.	"Engineering Ethics includes Human Values" by M.Govindarajan, S.Natarajan- and, V.S.Senthil Kumar-PHI Learning Pvt Ltd-2009.
2.	"Engineering Ethics" by Harris, Pritchard and Rabins, CENGAGE Learning, India Edition, 2009.
3.	"Ethics in Engineering" by Mike W. Martin and Roland Schinzinger-Tata McGraw-Hill-2003.
4.	"Professional Ethics and Morals" by Prof.A.R.Aryasri, DhanikotaSuyodhana-Maruthi Publications.
5.	"Professional Ethics and Human Values" by A.Alavudeen, R.Kalil Rahman and M.Jayakumaran-LaxmiPublications.
6.	"Professional Ethics and Human Values" by Prof.D.R.Kiran
7.	"Indian Culture, Values and Professional Ethics" by PSR Murthy- BS Publication.
8.	Professional Ethics by R.Subramaniam - Oxford publications, New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC1203	**MC**	**--**	**--**	**2**	**--**	**--**	**--**	**--**

NATIONAL SERVICE SCHEME(NSS)

(Common to All Branches)

Course Objectives:

1.	To understand the community and understand themselves in relation to their community.
2.	Identify the needs and problems of the community and involve them in problem solving process.
3.	Utilize their knowledge for finding practical solution to individual and community problems.

Course Outcomes: Student will be able to

S.No		Knowledge Level
1.	**understand** general orientation about community service, voluntarism role and responsibility of NSS volunteer.	K2
2.	**Analyze** about the community he live in.	K4
3.	**Asses** the life in adopted villages.	K5
4.	**Identify** the importance of national days and attain participation in it.	K3

SYLLABUS

1.	Volunteerism- community and beyond(Theory).
2.	Role and responsibility of NSS volunteer (Theory).
3.	General orientation about community service(Theory).
4.	Arranging lectures on social issues in schools or villages(Theory).
5.	Arranging rally's on social issues.
6.	Socio economic survey in adopted villages
7.	Plantation of saplings.
8.	Blood donation camp
9.	Rainwater harvesting awareness camp.
10.	Celebration of national days as per NSS list.